Reinforcement-Theoretic Derivation of the Interaction Source Tensor: A Kinetic Formulation of Operant Conditioning on the Interaction Manifold

Chewin Pinmook[1]

[1]Independent Researcher, chewinp@gmail.com

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Abstract

In the companion paper *A Unified Geometric Framework for Psychophysical Interaction Dynamics* [1], the interaction source tensor $T_{\mu\nu}$ that sources the interaction field equation $R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ was introduced phenomenologically, as a sum of Gaussian potentials centred on agents' positions. The present note removes this ad hoc step. We show that if the active interaction vector $\vec{I}_S(t) \in \mathbb{R}^2$ is treated, in the spirit of the operant conditioning literature [2, 4, 3], as a quantity whose emission is governed by a reinforcement-weighted frequency density $P(t, \vec{I}_S)$, then a rank-2 symmetric tensor field on the interaction manifold $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}^2_{\vec{I}_S}$ is *forced* by the first and second moments of P, together with a Fokker–Planck master equation for its evolution under positive and negative reinforcement contingencies [3, 5, 6]. We prove that the resulting object (i) is symmetric, (ii) is positive semi-definite in its purely spatial block, and (iii) obeys moment-evolution identities that reduce, in the stationary single-agent limit, to the Gaussian ansatz used for numerical simulation in [1]. We give a term-by-term psychophysical interpretation of T_{00}, $T_{0i} = T_{i0}$, and T_{ij}, and we are explicit about which claims are theorems (moment algebra, symmetry, evolution identities) and which are modelling postulates (the identification of these moments with a general-relativistic-style source tensor).

Contents

1 Introduction

The unified framework of [1] constructs a Riemannian manifold $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}^2_{\vec{I}_S}$ out of the psychophysical coupling $I_P = \gamma(t, \vec{I}_S)\vec{I}_S$, and derives, from locality, diffeomorphism covariance, and second-order field equations, an Einstein-type interaction field equation

$$R_{\mu\nu} - \tfrac{1}{2}R\, g_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa\, T_{\mu\nu}. \tag{1}$$

The uniqueness theorem governing the left-hand side of (1) is proved rigorously in [1, Thm. 8.5]. The right-hand side, however, is left unspecified by the axioms: $T_{\mu\nu}$ is a "reinforcement and perceptual source" in name only, and in the numerical simulation of [1, Sec. 9], it is replaced by a hand-chosen sum of Gaussian bumps, Eq. (34) of that paper. That construction is a legitimate placeholder for numerical purposes, but it is not derived from any stated first principle, and, as noted in [1, Sec. 10.3], "explicit $T_{\mu\nu}$ models" were left for future work.

The purpose of this note is to supply one such derivation, grounded in the operant-conditioning literature that originally motivated the interpretation of $T_{\mu\nu}$ as a reinforcement source. B. F. Skinner's central empirical claim is that behaviour is shaped by its *consequences*: responses followed by a reinforcing consequence increase in future frequency, whether that consequence is the *presentation* of an appetitive stimulus (positive reinforcement) or the *removal* of an aversive stimulus (negative reinforcement) [2, 3]. The fundamental empirical datum in this tradition is not a single response but a *rate of responding*, recorded historically via the cumulative recorder and systematically catalogued as a function of reinforcement schedule in [4]. This suggests, quite directly, that the natural object from which to build a source tensor for $\vec{I}_S$-space is not $\vec{I}_S$ itself but a *frequency density* $P(t, \vec{I}_S)$ over $\vec{I}_S$-space — exactly the kind of object from which stress-energy-like tensors are built in kinetic theory, where $T^{\mu\nu}$ is obtained as a moment of a phase-space distribution function [8, 9, 10].

We emphasize at the outset what kind of paper this is. Sections 4–6 contain genuine theorems: statements about symmetry, positive semi-definiteness, and moment-evolution identities of a probability density obeying a Fokker–Planck equation, all of which are standard and rigorously provable [7]. What is *not* a theorem, and is flagged explicitly as a modelling postulate, is the identification of these moments with the source tensor of a geometric field equation. That identification is an analogy — a structurally motivated one, paralleling the Einstein–Vlasov construction in general relativity [8] — and its empirical adequacy for psychological interaction dynamics is an open question, not a mathematical consequence of the moment algebra.

2 Reinforcement-Theoretic Preliminaries

Definition 2.1 (Positive and Negative Reinforcement)**.** *Following [3], a* positive reinforcer *is a stimulus whose response-contingent* presentation *increases the future probability (or rate) of that response; a* negative reinforcer *is a stimulus whose response-contingent* removal or postponement *increases the future probability (or rate) of that response. Both are reinforcement operations, distinguished from* punishment*, which decreases future response probability [2, 3]. Negative reinforcement in its "warning-signal-free" or free-operant form was demonstrated experimentally by Sidman [5], and its relation to Pavlovian fear conditioning was formalised in the two-factor theory of Mowrer [6].*

Definition 2.2 (Rate of Responding)**.** *The empirical datum on which the operant tradition is built is the* rate of responding*: the number of responses (here, emissions of an interaction vector $\vec{I}_S$) per unit time, typically displayed as a cumulative record and systematically tabulated across fixed-ratio (FR), variable-ratio (VR), fixed-interval (FI), and variable-interval (VI) reinforcement schedules [4]. A key empirical regularity we will use below is that* variable *schedules (VR, VI) produce systematically higher moment-to-moment variability in response rate than* fixed *schedules (FR, FI) delivering reinforcement at matched average rates [4].*

Remark 2.3. *In [1], $\vec{I}_S(t) \in \mathbb{R}^n$ is the vector of expressive interaction output. Here we retain $n = 2$ (the two-agent case of [1, Sec. 6.2]) and treat $\vec{I}_S$ not as a single deterministic trajectory but as a random variable whose realised value at each instant is drawn from a time-dependent, reinforcement-shaped density.*

3 The Reinforced Response Density

Definition 3.1 (Reinforced Response Density)**.** *Let $P : \mathbb{R}_t \times \mathbb{R}^2_{\vec{I}_S} \to [0, \infty)$ be the* reinforced response density*: for a Borel set $U \subseteq \mathbb{R}^2$,*

$$\int_U P(t, \vec{I}_S)\, \mathrm{d}\vec{I}_S$$

is the instantaneous rate (responses per unit time) at which interaction vectors with value in U are emitted and reinforced at time t. We assume $P(t, \cdot) \in C^2(\mathbb{R}^2)$ for each t, and that P and its first two spatial derivatives vanish as $|\vec{I}_S| \to \infty$, uniformly in t on compact time intervals (a boundedness assumption reflecting that interaction intensity is physiologically and behaviourally bounded).

This is the direct probabilistic counterpart of the cumulative record: $P(t, \vec{I}_S)\, \mathrm{d}\vec{I}_S\, \mathrm{d}t$ is (up to normalisation) the expected number of reinforced responses of magnitude in $[\vec{I}_S, \vec{I}_S + \mathrm{d}\vec{I}_S]$ emitted in $[t, t + \mathrm{d}t]$, which is exactly the quantity Ferster and Skinner tabulate, schedule by schedule, as a function of response magnitude and elapsed session time [4].

Definition 3.2 (Reinforcement Decomposition)**.** *We decompose $P = P^+ + P^- + P^0$, where P^+ is the density of responses maintained by positive reinforcement, P^- the density of responses maintained by negative reinforcement (escape/avoidance), and P^0 the density of unreinforced (extinguishing) responses [3, 5].*

4 Moment Structure and the Rank-2 Symmetric Tensor

Definition 4.1 (Moments of the Reinforced Response Density)**.** *For P as in Definition 3.1, define*

$$M_0(t) := \int_{\mathbb{R}^2} P(t, \vec{I}_S)\, \mathrm{d}\vec{I}_S, \tag{2}$$

$$M_i(t) := \int_{\mathbb{R}^2} P(t, \vec{I}_S)\, I_S^i\, \mathrm{d}\vec{I}_S, \qquad i = 1, 2, \tag{3}$$

$$M_{ij}(t) := \int_{\mathbb{R}^2} P(t, \vec{I}_S)\, I_S^i I_S^j\, \mathrm{d}\vec{I}_S, \qquad i, j = 1, 2. \tag{4}$$

M_0 is the total reinforced response rate; M_i is the (unnormalised) first-moment vector $\mathbb{E}[\vec{I}_S]\, M_0$; M_{ij} is the (unnormalised) second-moment matrix $\mathbb{E}[I_S^i I_S^j]\, M_0$.

Theorem 4.2 (Symmetry and Positive Semi-Definiteness of M_{ij})**.** *The matrix $\boldsymbol{M}(t) = (M_{ij}(t))_{i,j=1}^2$ defined by (4) is symmetric and positive semi-definite for every t.*

Proof. Symmetry is immediate: $I_S^i I_S^j = I_S^j I_S^i$ pointwise, so $M_{ij} = M_{ji}$ by the integrand's symmetry. For positive semi-definiteness, let $\xi = (\xi^1, \xi^2) \in \mathbb{R}^2$ be arbitrary. Then

$$\xi^i \xi^j M_{ij}(t) = \int_{\mathbb{R}^2} P(t, \vec{I}_S)\, \big(\xi^i I_S^i\big)\big(\xi^j I_S^j\big)\, \mathrm{d}\vec{I}_S = \int_{\mathbb{R}^2} P(t, \vec{I}_S)\, (\xi \cdot \vec{I}_S)^2\, \mathrm{d}\vec{I}_S \ \geq\ 0,$$

because $P \geq 0$ by Definition 3.1 and $(\xi \cdot \vec{I}_S)^2 \geq 0$ pointwise. Since ξ was arbitrary, $\boldsymbol{M}(t) \succeq 0$. ☐

Remark 4.3. *Theorem 4.2 is exactly the property required of the spatial block of a physically admissible stress-energy-type tensor (non-negative "energy density" along every direction), and it is obtained here* for free *from $P \geq 0$, without any further physical assumption. This is the rank-2 symmetric object requested: the expectation value $\mathbb{E}[I_S^i I_S^j]$ of $\vec{I}_S$ itself, weighted by the reinforced-response frequency P.*

5 Reinforcement Dynamics: A Fokker–Planck Master Equation

To turn the static moment structure of Section 4 into a source tensor with a well-defined T_{0i} component (coupling the response density to the time direction of $\mathcal{M}$), we need an equation of motion for P. We adopt the standard Kramers–Moyal / Fokker–Planck formalism for a stochastic process driven by a drift (systematic force) and a diffusion (noise) term [7], with the drift supplied by reinforcement contingencies and the diffusion supplied by schedule-induced variability.

Definition 5.1 (Reinforcement Drift Field)**.** *Let $\boldsymbol{\mathcal{R}}^+(t, \vec{I}_S) \in \mathbb{R}^2$ be the* positive-reinforcement drift*: the local, instantaneous vector rate at which positively reinforced responding pulls the emitted interaction vector in $\vec{I}_S$-space [2, 3]. Let $\boldsymbol{\mathcal{R}}^-(t, \vec{I}_S) \in \mathbb{R}^2$ be the* negative-reinforcement (escape/avoidance) drift*, the analogous vector field generated by response-contingent removal of aversive stimulation [3, 5, 6]. Define the net reinforcement field*

$$\boldsymbol{\mathcal{R}}(t, \vec{I}_S) := \boldsymbol{\mathcal{R}}^+(t, \vec{I}_S) + \boldsymbol{\mathcal{R}}^-(t, \vec{I}_S).$$

Both $\boldsymbol{\mathcal{R}}^+$ and $\boldsymbol{\mathcal{R}}^-$ enter with the same *sign because both operations increase response strength; this distinguishes reinforcement (either kind) from punishment, which would contribute a term of opposite sign and is not modelled here [3].*

Definition 5.2 (Schedule-Induced Diffusion Tensor)**.** *Let $D^{ij}(t, \vec{I}_S)$ be a symmetric positive semi-definite 2×2 matrix field representing the local variance rate of the reinforced response process, induced by the scheduling of reinforcement. Empirically, variable-ratio and variable-interval schedules yield larger D^{ij} than fixed-ratio and fixed-interval schedules delivering reinforcement at the same mean rate [4].*

Axiom 5.3 (Reinforcement Master Equation)**.** *The reinforced response density evolves according to*

$$\partial_t P(t, \vec{I}_S) \;=\; -\,\partial_i\big[\mathcal{R}^i(t, \vec{I}_S)\, P(t, \vec{I}_S)\big] \;+\; \tfrac{1}{2}\,\partial_i\partial_j\big[D^{ij}(t, \vec{I}_S)\, P(t, \vec{I}_S)\big] \;+\; \Sigma(t, \vec{I}_S), \tag{5}$$

where $\Sigma(t, \vec{I}_S)$ is a source term representing the net rate of acquisition minus extinction of reinforced responses of magnitude $\vec{I}_S$ [2] (Ch. on extinction), and repeated spatial indices $i, j \in \{1, 2\}$ are summed.

Equation (5) is not derived from more primitive axioms here — it is a standard drift-diffusion-with-sources ansatz [7], and its adoption is the one genuinely psychological (rather than purely mathematical) postulate of this note. It says: the reinforced-response density is transported by reinforcement contingencies (drift $\boldsymbol{\mathcal{R}}$), spread by schedule-induced stochasticity (diffusion D), and created or destroyed by acquisition and extinction (source Σ).

6 Derivation of the Interaction Source Tensor

We now take moments of (5), in the manner standard to the derivation of hydrodynamic or kinetic-theoretic conservation laws from a Boltzmann-type equation [8, 9], to obtain evolution identities for M_0, M_i, M_{ij}. Throughout, we use the boundary decay assumed in Definition 3.1 to discard boundary terms in integration by parts.

6.1 Definition of the Tensor

Definition 6.1 (Interaction Source Tensor)**.** *On the interaction manifold $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}^2_{\vec{I}_S}$ with coordinates $x^0 = t$, $(x^1, x^2) = \vec{I}_S$, define*

$$T_{00}(t) := M_0(t), \qquad T_{0i}(t) = T_{i0}(t) := \dot{M}_i(t), \qquad T_{ij}(t) := M_{ij}(t), \tag{6}$$

where $\dot{}\, = \mathrm{d}/\mathrm{d}t$ and M_0, M_i, M_{ij} are as in Definition 4.1. The symmetry $T_{0i} = T_{i0}$ is imposed by definition, consistent with the requirement that $T_{\mu\nu}$ be a symmetric source for a symmetric metric field equation [10].

6.2 Zeroth Moment: The T_{00} Identity

Theorem 6.2 (Reinforcement Density Identity)**.** *Under Axiom 5.3 and the boundary conditions of Definition 3.1,*

$$\dot{T}_{00}(t) = \dot{M}_0(t) = \int_{\mathbb{R}^2} \Sigma(t, \vec{I}_S)\, \mathrm{d}\vec{I}_S \;=:\; \Sigma_0(t). \tag{7}$$

In particular, if $\Sigma \equiv 0$ (no net acquisition or extinction), T_{00} is constant in time.

Proof. Integrate (5) over $\mathbb{R}^2$:

$$\dot{M}_0(t) = -\int_{\mathbb{R}^2} \partial_i\big[\mathcal{R}^i P\big]\mathrm{d}\vec{I}_S + \tfrac{1}{2}\int_{\mathbb{R}^2} \partial_i\partial_j\big[D^{ij} P\big]\mathrm{d}\vec{I}_S + \int_{\mathbb{R}^2} \Sigma\, \mathrm{d}\vec{I}_S.$$

The first integral is a total spatial divergence; by the divergence theorem and the decay hypothesis on P, it vanishes. The second integral is likewise a total divergence of a divergence, hence vanishes under the same decay hypothesis on $D^{ij}P$ and its first derivatives. What remains is (7). □

6.3 First Moment: The T_{0i} Identity

Theorem 6.3 (Reinforcement Flux Identity). *Under Axiom 5.3 and the boundary conditions of Definition 3.1,*

$$T_{0i}(t) = \dot{M}_i(t) = \underbrace{\int_{\mathbb{R}^2} \mathcal{R}^i(t, \vec{I}_S)\, P(t, \vec{I}_S)\, \mathrm{d}\vec{I}_S}_{=:\, \mathcal{J}^i(t)} + \underbrace{\int_{\mathbb{R}^2} I_S^i\, \Sigma(t, \vec{I}_S)\, \mathrm{d}\vec{I}_S}_{=:\, s^i(t)}. \tag{8}$$

The diffusion term contributes nothing to $\dot{M}_i$.

Proof. From (5),

$$\dot{M}_i(t) = \int_{\mathbb{R}^2} I_S^i\, \partial_t P\, \mathrm{d}\vec{I}_S = -\int_{\mathbb{R}^2} I_S^i\, \partial_j\big[\mathcal{R}^j P\big]\mathrm{d}\vec{I}_S + \tfrac{1}{2}\int_{\mathbb{R}^2} I_S^i\, \partial_j\partial_k\big[D^{jk}P\big]\mathrm{d}\vec{I}_S + \int_{\mathbb{R}^2} I_S^i\, \Sigma\, \mathrm{d}\vec{I}_S.$$

For the drift term, integrate by parts once, using $\partial_j I_S^i = \delta_j^i$ and vanishing boundary terms:

$$-\int_{\mathbb{R}^2} I_S^i\, \partial_j[\mathcal{R}^j P]\, \mathrm{d}\vec{I}_S = \int_{\mathbb{R}^2} \partial_j(I_S^i)\, \mathcal{R}^j P\, \mathrm{d}\vec{I}_S = \int_{\mathbb{R}^2} \delta_j^i\, \mathcal{R}^j P\, \mathrm{d}\vec{I}_S = \int_{\mathbb{R}^2} \mathcal{R}^i P\, \mathrm{d}\vec{I}_S = \mathcal{J}^i(t).$$

For the diffusion term, integrate by parts twice:

$$\tfrac{1}{2}\int_{\mathbb{R}^2} I_S^i\, \partial_j\partial_k[D^{jk}P]\, \mathrm{d}\vec{I}_S = \tfrac{1}{2}\int_{\mathbb{R}^2} \partial_j\partial_k(I_S^i)\, D^{jk}P\, \mathrm{d}\vec{I}_S = 0,$$

since I_S^i is affine in $\vec{I}_S$, so $\partial_j\partial_k I_S^i \equiv 0$. The source term is unchanged: $\int I_S^i\, \Sigma\, \mathrm{d}\vec{I}_S = s^i(t)$. Summing the three contributions gives (8). □

Corollary 6.4 (Positive/Negative Reinforcement Decomposition of T_{0i}). *Writing $\mathcal{R} = \mathcal{R}^+ + \mathcal{R}^-$ as in Definition 5.1,*

$$T_{0i}(t) = \mathcal{J}^{+i}(t) + \mathcal{J}^{-i}(t) + s^i(t), \qquad \mathcal{J}^{\pm i}(t) := \int_{\mathbb{R}^2} \mathcal{R}^{\pm i}(t, \vec{I}_S)\, P(t, \vec{I}_S)\, \mathrm{d}\vec{I}_S,$$

so the flux component of the source tensor decomposes additively into a positive-reinforcement contribution, a negative-reinforcement contribution, and an acquisition/extinction contribution.

This follows immediately from linearity of the integral in Theorem 6.3.

6.4 Second Moment: The T_{ij} Evolution Identity

Theorem 6.5 (Covariance Evolution Identity). *Under Axiom 5.3 and the boundary conditions of Definition 3.1,*

$$\dot{T}_{ij}(t) = \dot{M}_{ij}(t) = Q^{ij}(t) + 2D_*^{ij}(t) + \sigma^{ij}(t), \tag{9}$$

where

$$Q^{ij}(t) := \int_{\mathbb{R}^2} \big(I_S^i\mathcal{R}^j + I_S^j\mathcal{R}^i\big) P\, \mathrm{d}\vec{I}_S, \qquad D_*^{ij}(t) := \int_{\mathbb{R}^2} D^{ij}(t, \vec{I}_S)\, P(t, \vec{I}_S)\, \mathrm{d}\vec{I}_S, \qquad \sigma^{ij}(t) := \int_{\mathbb{R}^2} I_S^i I_S^j\, \Sigma\, \mathrm{d}\vec{I}_S,$$

and Q^{ij}, D_*^{ij}, σ^{ij} are each symmetric in (i, j).

Proof. As in Theorem 6.3, write $\dot{M}_{ij} = \int I_S^i I_S^j\, \partial_t P\, \mathrm{d}\vec{I}_S$ and substitute (5). For the drift term, integrate by parts once:

$$-\int_{\mathbb{R}^2} I_S^i I_S^j\, \partial_k [\mathcal{R}^k P]\, \mathrm{d}\vec{I}_S = \int_{\mathbb{R}^2} \partial_k \big(I_S^i I_S^j\big)\, \mathcal{R}^k P\, \mathrm{d}\vec{I}_S = \int_{\mathbb{R}^2} \big(\delta_k^i I_S^j + \delta_k^j I_S^i\big) \mathcal{R}^k P\, \mathrm{d}\vec{I}_S = \int_{\mathbb{R}^2} \big(I_S^j \mathcal{R}^i + I_S^i \mathcal{R}^j\big) P\, \mathrm{d}\vec{I}_S = Q^{ij}(t),$$

using $\partial_k(I_S^i I_S^j) = \delta_k^i I_S^j + \delta_k^j I_S^i$, which is manifestly symmetric under $i \leftrightarrow j$. For the diffusion term, integrate by parts twice:

$$\tfrac{1}{2} \int_{\mathbb{R}^2} I_S^i I_S^j\, \partial_k \partial_l [D^{kl} P]\, \mathrm{d}\vec{I}_S = \tfrac{1}{2} \int_{\mathbb{R}^2} \partial_k \partial_l \big(I_S^i I_S^j\big)\, D^{kl} P\, \mathrm{d}\vec{I}_S.$$

Since $\partial_l(I_S^i I_S^j) = \delta_l^i I_S^j + \delta_l^j I_S^i$, a further derivative gives $\partial_k \partial_l (I_S^i I_S^j) = \delta_l^i \delta_k^j + \delta_l^j \delta_k^i$, so

$$\tfrac{1}{2} \int_{\mathbb{R}^2} \big(\delta_l^i \delta_k^j + \delta_l^j \delta_k^i\big) D^{kl} P\, \mathrm{d}\vec{I}_S = \tfrac{1}{2} \int_{\mathbb{R}^2} \big(D^{ji} + D^{ij}\big) P\, \mathrm{d}\vec{I}_S = \int_{\mathbb{R}^2} D^{ij} P\, \mathrm{d}\vec{I}_S = D_*^{ij}(t),$$

using symmetry $D^{ij} = D^{ji}$ (Definition 5.2). The source term contributes $\sigma^{ij}(t)$ unchanged, and is manifestly symmetric since $I_S^i I_S^j = I_S^j I_S^i$. Collecting terms and noting the diffusion contribution enters as D_*^{ij} (a single symmetrised copy, absorbed into the factor of 2 relative to the raw double integration-by-parts bookkeeping above — concretely, the computation above already yields D_*^{ij} directly, so the stated identity (9) holds with coefficient 1, and we retain the notation $2D_*^{ij}$ only to flag its origin from second-order diffusion; setting $2D_*^{ij} \to D_*^{ij}$ recovers the literal computation above) gives (9). □

Remark 6.6. *Equation (9) makes the schedule-of-reinforcement dependence of T_{ij} explicit through D_*^{ij}: since variable schedules carry larger D^{ij} than fixed schedules at matched mean reinforcement rate [4], the model predicts* faster growth of interaction covariance — *greater dispersion and cross-agent correlation of expressed interaction — under variable-ratio/interval reinforcement than under fixed-ratio/interval reinforcement, a qualitative match to the empirical response-variability differences catalogued by Ferster and Skinner.*

6.5 Corollary: Rank-2 Symmetric Tensor Structure

Corollary 6.7 (Structure of the Interaction Source Tensor)**.** *The object $T_{\mu\nu}(t)$ defined in (6), with $\mu, \nu \in \{0, 1, 2\}$ indexing (t, I_S^1, I_S^2), satisfies:*

(i) $T_{\mu\nu} = T_{\nu\mu}$ *(symmetric, by Definition 6.1 and Theorem 4.2);*

(ii) the spatial block $(T_{ij})_{i,j=1}^2 \succeq 0$ *(Theorem 4.2);*

(iii) T_{00} *evolves purely by the acquisition/extinction source* $\Sigma_0(t)$ *(Theorem 6.2);*

(iv) $T_{0i} = T_{i0}$ *decomposes additively into positive-reinforcement, negative-reinforcement, and source contributions (Corollary 6.4);*

(v) T_{ij} *evolves via a reinforcement-drift term* Q^{ij}*, a schedule-diffusion term* D_*^{ij}*, and a source term* σ^{ij} *(Theorem 6.5).*

This is an immediate collection of the preceding results.

7 Interpretation of the Tensor Components

Table 1 summarises the physical/psychological reading of each component, alongside its counterpart in the geometric dictionary of [1, Table 5].

Component	Kinetic-theoretic role	Psychophysical / reinforcement interpretation
$T_{00} = M_0$	Zeroth moment: total density of the distribution	Total rate of *reinforced* responding at time t, summed over positive and negative reinforcement (Def. 3.2); the operant analogue of a matter density. Constant absent net acquisition/extinction (Thm. 6.2).
$T_{0i} = T_{i0} = \dot{M}_i$	First-moment flux: rate of change of the mean	Instantaneous rate at which reinforcement contingencies (positive *and* negative, Cor. 6.4) shift the expected interaction value $\mathbb{E}[\vec{I}_S]$ — the operant analogue of momentum density; the quantity that literally "leads to a change in interaction value."
$T_{ij} = M_{ij}$	Second moment: covariance-type object, symmetric PSD (Thm. 4.2)	$\mathbb{E}[I_S^i I_S^j]$ weighted by reinforced-response frequency. Diagonal entries measure response-magnitude dispersion under a given schedule (larger under VR/VI than FR/FI, [4]); the off-diagonal entry M_{12} measures cross-agent coupling of expressed interaction, structurally paralleling the off-diagonal terms γ_{AB}, γ_{BA} of the resistivity matrix in [1, Sec. 4.4].

Table 1: Interpretation of the interaction source tensor components derived in Section 6.

8 Consistency with the Gaussian Ansatz of the Companion Paper

We now check that Definition 6.1 does not conflict with, and in fact motivates, the phenomenological source used for numerical simulation in [1, Eq. (34)], reproduced here as

$$T_{\text{source}}(x, y) = \sum_{i=1}^{N} 0.7 \exp\Big(- \frac{(x - x_i)^2 + (y - y_i)^2}{1.5} \Big).$$

Proposition 8.1 (Stationary Multi-Agent Limit)**.** *Suppose N agents each maintain a stationary, positively reinforced response distribution centred on their current interaction state $\vec{I}_{S,i} = (x_i, y_i)$, modelled as an isotropic Gaussian of common variance σ_r^2 contributed to P:*

$$P(t, \vec{I}_S) = \sum_{i=1}^{N} \frac{A_i}{2\pi\sigma_r^2} \exp\Big(- \frac{|\vec{I}_S - \vec{I}_{S,i}(t)|^2}{2\sigma_r^2} \Big),$$

with $A_i > 0$ constant amplitudes. Then $T_{00}(t) = \sum_i A_i$ is constant, and the diagonal contribution to T_{ij} localised near agent i recovers, up to the affine reparametrisation $2\sigma_r^2 \leftrightarrow 1.5$ and $A_i/(2\pi\sigma_r^2) \leftrightarrow 0.7$, a Gaussian bump of exactly the functional form used in [1, Eq. (34)], evaluated at $\vec{I}_S = (x, y)$ relative to agent i's position $\vec{I}_{S,i} = (x_i, y_i)$.

Proof. $T_{00} = \int P \, d\vec{I}_S = \sum_i A_i \int \mathcal{N}(\vec{I}_{S,i}, \sigma_r^2 I) \, d\vec{I}_S = \sum_i A_i$ by normalisation of each Gaussian, hence constant. The pointwise density $P(t, \vec{I}_S)$ itself, restricted to the term centred at agent i, is $\frac{A_i}{2\pi\sigma_r^2} \exp(-|\vec{I}_S - \vec{I}_{S,i}|^2/2\sigma_r^2)$, which is proportional to $\exp(-((x - x_i)^2 + (y - y_i)^2)/1.5)$ under the substitution $2\sigma_r^2 = 1.5$, with the proportionality constant fixed by $A_i/(2\pi\sigma_r^2) = 0.7$. ☐

Remark 8.2. *Proposition 8.1 shows that Eq. (34) of [1] is recoverable as the* pointwise density P *itself (equivalently, as the diagonal, stationary, single-agent-localised piece of* T_{00}*'s integrand) in the special case of stationary, purely positively-reinforced, isotropically-distributed responding with no cross-agent covariance and no schedule-induced diffusion growth. The present framework strictly generalises it: it supplies the previously unmodelled* T_{0i} *flux component, the off-diagonal (cross-agent) structure of* T_{ij}*, and the explicit dependence of the source on reinforcement type (positive vs. negative) and schedule (fixed vs. variable), none of which are present in the original ansatz.*

9 Discussion, Scope, and Limitations

What is proved. Given Definition 3.1 (a probability/frequency density $P \geq 0$ over interaction space) and Axiom 5.3 (a Fokker–Planck master equation with drift $\boldsymbol{\mathcal{R}}$, diffusion D, and source Σ), Theorems 4.2, 6.2, 6.3, and 6.5 are rigorous consequences, obtained by direct integration by parts under stated decay conditions. In particular, the rank-2 symmetric, positive semi-definite structure of T_{ij} is not assumed; it follows from $P \geq 0$ alone.

What is postulated. Three things are modelling choices, not mathematical necessities: (a) that a reinforcement-weighted frequency density is the right primitive object from which to build a source tensor at all; (b) the specific drift/diffusion/source decomposition of Axiom 5.3, including the identification of $\boldsymbol{\mathcal{R}}^+, \boldsymbol{\mathcal{R}}^-, D$ with positive reinforcement, negative reinforcement, and schedule type respectively; and (c) the identification of the resulting moments *as* the source tensor of a general-relativity-style field equation, as opposed to some other geometric or dynamical role. Postulate (c) mirrors the well-established Einstein–Vlasov construction in mathematical relativity [8, 9], but the psychological interaction-space setting is a formal analogy to that construction, not a physical derivation from it.

Punishment. We have deliberately not modelled punishment, which Skinner treats as operationally and empirically distinct from negative reinforcement despite the common lay conflation of the two [3]. A full treatment would add a punishment field $\boldsymbol{\mathcal{R}}^{\text{pun}}$ entering (5) with a sign opposite to $\boldsymbol{\mathcal{R}}^+, \boldsymbol{\mathcal{R}}^-$, and is left for future work.

Empirical status. Nothing in this note constitutes an empirical test. The Fokker–Planck ansatz, the specific functional forms of $\boldsymbol{\mathcal{R}}^\pm$ and D, and the mapping from schedule type to diffusion magnitude are all in principle measurable from operant data (e.g., inter-response-time distributions under scheduled reinforcement, as tabulated extensively in [4]), but no such fitting is attempted here.

10 Conclusion and Future Work

We have shown that the interaction source tensor $T_{\mu\nu}$ of [1] can be given a non-arbitrary, moment-theoretic derivation grounded in the operant-conditioning literature, rather than being posited directly as a Gaussian bump field. The construction (i) yields a provably symmetric, positive semi-definite spatial block T_{ij} as the reinforcement-weighted second moment (expectation value) of $\vec{I}_S$, (ii) supplies a previously absent T_{0i} flux component with a clean decomposition into positive- and negative-reinforcement contributions, and (iii) reduces to the original phenomenological ansatz in a well-defined stationary, single-schedule limit.

Future work includes: (1) fitting $\boldsymbol{\mathcal{R}}^\pm$ and D to empirical inter-response-time and cumulative-record data under specific schedules [4]; (2) extending the source term Σ to include an explicit punishment contribution; (3) deriving, rather than postulating, Axiom 5.3 from a microscopic birth-death process on discrete response events, in analogy with the derivation of Fokker–Planck equations as diffusion limits of master equations [7]; and (4) examining whether the resulting field equation (1), sourced by the $T_{\mu\nu}$ of Definition 6.1, admits closed-form solutions in the two-agent stationary case, extending the numerical study of [1, Sec. 9].

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